When Does "Thinking" Help a Model See?

Investigating Latent Computation in Multimodal Reasoning

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Vision-Language Research Group · `MTok Project`

Why Vision Reasoning?

Three Canonical Failure Modes

1. **Jigsaw Puzzle**: Which piece fills the vacant gap?

   Requires mentally rotating, inverting, and fitting 2D boundaries.

2. **Relative Depth**: Which marked point is closer to camera?

   Requires 3D geometry and occlusion reasoning from 2D pixel cues.

3. **Perspective-Taking**: Facing fridge at X, is table front-left?

   Requires dynamic coordinate frame re-anchoring.

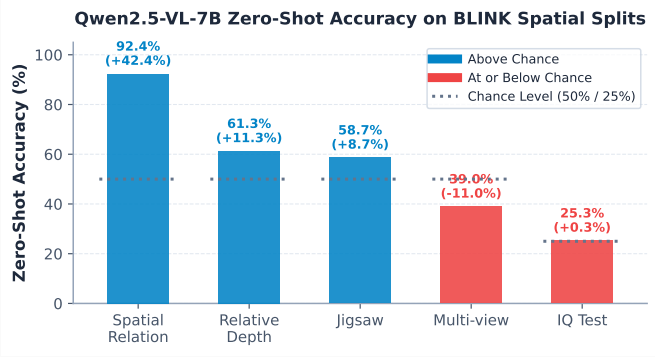
Perception $\neq$ Reasoning

A model that can label every object in a scene **still fails** these tasks.

Central Working Claim:
Language is a lossy channel for spatial computation.

Spatial structure resists verbal decomposition; verbalising it forces unnatural sequential commitments.

The Spatial Gap in One Chart



Key Empirical Reality

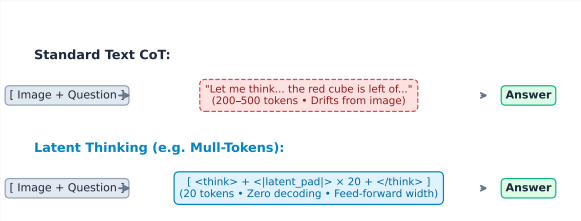
- **Perception is Solved**: Spatial Relation = **92.4%** *(+42.4% over chance)*
- **Manipulation Collapses**: Multi-View = **39.0%** *(vs. 50% chance)*
  IQ Test = **25.3%** *(vs. 25% chance)*
- **Why Qwen2.5-VL**: Standard open literature anchor for spatial VLM research.

Four Paradigms for Visual Reasoning

Paradigm	Core Mechanism	Representative Work	Cost / Primary Failure
Text CoT	Write reasoning out in words	Wei et al. (2022); Video-R1	Drifts from image; **hurts vs. direct answer** on spatial benchmarks (~ -3 pp)
Tools / Sketch	External cropper, detector, sketchpad	Visual Sketchpad (2024); DeepEyes; Pixel Reasoner	Brittle, external modules, narrow manipulation vocabulary
Gen Images	Produce intermediate pictures	Unified models (Chameleon, Blip3-o)	Extremely expensive to train and run
Latent Tokens	Reason in continuous hidden states	COCONUT (2024); Mirage; LVR; Monet; **Mull-Tokens**	Uninterpretable: **does latent computation actually occur?**

The Core Hinge: If verbalising spatial manipulation actively harms performance, the logical move is to reason **without writing it down** $\rightarrow$ **Latent Tokens**.

The Latent Reasoning Concept



Intellectual Lineage

- **Pause Tokens** (Goyal et al., ICLR 2024): Inserting content-free tokens grants extra decoder compute.
- **COCONUT** (Hao et al., 2024): Feeds hidden states back as inputs rather than decoding to words.

The Multimodal Hypothesis

Spatial computation never had a good verbal form in the first place.
→ The bet in vision is that latent computation helps **more than in language** by bypassing verbal bottlenecks.

Evolution of Latent Multimodal Reasoning

2024: Foundations

- Pause Tokens (ICLR)
- COCONUT

Continuous latents demonstrated in language decoders.

2025: Multimodal

- Mirage
- Perception Tokens

Latents extended to image-text inputs.

Early 2026

- **Mull-Tokens**
- LVR / ILVR / Monet

Claiming double-digit gains on spatial tasks.

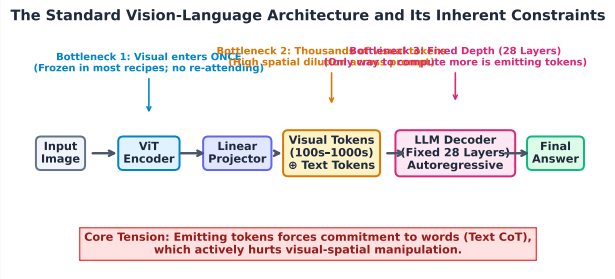
Late 2026: Audits

- Viveiros et al.
- Guo et al.
- Li et al., Zhang et al.

Are latents doing anything?

"There is now a set of papers asking whether any of this actually works, and I will come back to them once I have shown you my numbers."

The Standard Pipeline and Its Constraints



1. Single Input Pass

Frozen ViT encoder; visual tokens entered once. No mechanism to re-attend or re-inspect regions.

2. Fixed Depth

Decoder has fixed depth (28 layers); hard problems get identical sequential depth to easy ones.

3. Token Bottleneck

Extra compute requires emitting tokens → text CoT forces premature verbal commitment.

Four Concrete Limitations

1. Verbalisation Loses Spatial Structure

Text CoT underperforms direct answering by $\sim$ 3 points average on spatial benchmarks.

Evidence: Mull-Tokens Table 1, rows b vs. c.

2. Fixed Compute Budget

No architecture here dynamically allocates more sequential computation to harder items.

3. Visual Grounding Decays with Depth

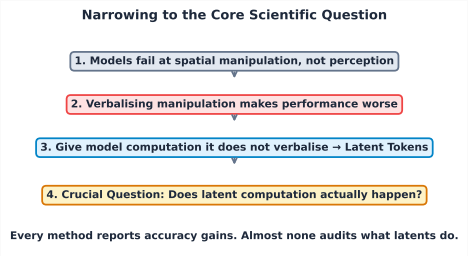
Reasoning progressively drifts from the image tokens; documented across the visual drift literature.

4. Latents are Uninterpretable

Latents remain uninterpretable black boxes.

The Mull-Tokens paper explicitly lists this as its own main limitation.

Narrowing to One Question: Competing Hypotheses



Hypothesis	Core Claim	Empirical Signature
Scratchpad	Carries image-derived computation	Perturbing latents hurts; counterfactual flips answer
Compute Buffer	Parallel compute; contents generic	Any trained filler works equally well
Format Key	Prefix triggers fine-tuned mode	**1 token works as well as twenty**

The Core Thesis: The paper claims the **Scratchpad**. Our empirical results point conclusively at the **Format Key**. That disagreement is where my project lives.

Mull-Tokens: The Minimal Testbed

Why Mull-Tokens?

- Minimalist instantiation of latent thinking.
- Modality-agnostic: uses uniform discrete tokens.
- No complex routing → cleanest mechanistic testbed.

Tokenizer Ground Truth

Confirmed via Qwen2.5-VL tokenizer: `<think>` and `</think>` are **plain text** (`<` + `think` + `>`), **not special tokens**. Mull adds no special boundary embeddings.

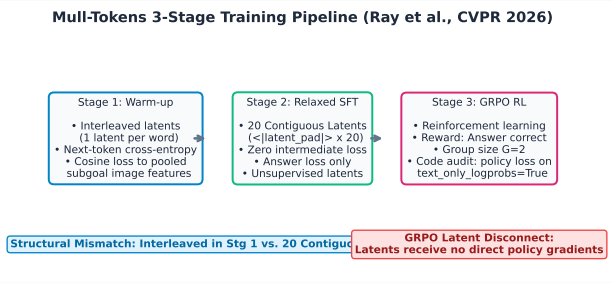
Inference-Time Intervention

Twenty copies of token ID 151665 prefilled into the assistant turn:

`<think>` + `<|latent_pad|>` × 20 + `</think>\n` → answer

- All 20 positions receive **identical input embeddings**.
- Whatever they compute must come from position and attention, not fed-in content.

Mull-Tokens 3-Stage Training Pipeline



Observation 1: Architectural Depth

Because latents are a fixed parallel prefix, latent k at layer ℓ only reads latents $< k$ at layer $\ell - 1$. Total serial compute is strictly 28 layers: **Mull adds width, not serial recurrence.**

Observation 2: Code Audit of Stage 3

Training config inspection reveals: `text_only_logprobs: True`. Policy gradients only backpropagate through answer tokens; **latents receive zero direct policy gradient!**

What the Paper Claims

Core Claim	Published Justification	Reported Metric
Beats Strongest Baseline	Outperforms direct-answer fine-tuning	**+5.40%** (DirAns: 61.40% → 66.80%)
Large Spatial Gain	Substantial surge on visual reasoning	**+15.34%** on BLINK Jigsaw (58.66% → 74.00%)
Inference Efficiency	20 latent tokens vs. verbose text CoT	**10–25× shorter** (20 vs. 200–500 tokens)
Multimodal Warm-up Vital	Cosine loss to subgoal image embeddings	**66.8%** vs. 65.9% (text) vs. 59.2% (no warm-up)
GRPO Latent Reinforcement	Latents made causally responsible	**+0.12%** overall (53.92% → 54.04%)

Read the last row out loud and pause: 50,000 GRPO iterations on 8×H100 GPUs yielded an overall gain of just 0.12 percentage points across all benchmarks.

My Critical Reading: Five Gaps

#	Identified Gap	Mechanistic Implication	Source
1	**No Causal Test**	Answer tokens can attend directly to image tokens, bypassing latents entirely; no intervention ruled this out.	Paper §3
2	**Confounded Data**	SAT data rose from 20% to 60% between Stage 1 & 2; gains may reflect domain adaptation, not latent thinking.	Appendix §6.1
3	**GRPO Does Not Reward Latents**	Released config sets `text_only_logprobs: True`; only final answer tokens enter the policy gradient.	**Source Code**
4	**Train/Test Mismatch**	Stage 1 interleaves 1 latent/word; Stage 2 injects 20 contiguous latents. Learned functional role may not transfer.	**Code** (Auth. Conf.)
5	**Spatial Collapse**	Subgoal images average-pooled to 1 vector, destroying 2D layout; method regresses 3.2 pp on Relative Depth.	Appendix

Transition: "I had five hypotheses about where this might be weak. The next part is what happened when I tested them."

Why Replicate First?

"A negative result is only believable if the positive result reproduces first."

Controlled Evaluation Hardware

- **GPU**: 1× NVIDIA A100-PCIE-40GB
- **OS / Env**: Ubuntu 20.04, CUDA 12.2
- **Software**: PyTorch 2.5.1, FlashAttention 2
- **Suite**: Hardened fork of `lmms-eval`

Strategic Validity Argument

- Pre-empts immediate challenge: *"Is your ablation setup broken?"*
- Validates checkpoint weights, prompt formatting, and tokenizer integrity.
- Fast iteration: Image suite runs in ~ 50 minutes on 1× A100.

Engineering the Testbed: Five Upstream Bugs Fixed

Upstream Code Defects Identified & Resolved

1. **Hardcoded Cluster Paths**: Task YAMLs referenced inaccessible internal cluster directories (`/home/jupyter/data_files/...`) → Rewired to HF streaming.
2. **Dataset Schema Mismatches**: SAT config naming, split errors, ERQA image key discrepancies (`images_encoded`), and checksum failures.
3. **Silent Chat-Template Corruption**: A line `text.replace("<|im_end|>\n", "")` stripped turn boundaries between **all** turns, silently collapsing system, user, and assistant into one invalid string.
4. **Broken Model Registration**: Hardcoded `sys.path.append("../models")` and missing `@register_model` decorators in evaluation harness.
5. **Logging Directory Crashes**: Hardcoded Google Cloud paths causing silent crashes.

Replication Verified Across 8 Benchmarks + Video

Benchmark Split (N)	Paper Claimed (SFT / GRPO)	Our Replication	Outcome vs. Paper
BLINK Spatial Relation (143)	81.81% / 81.81%	**85.31%**	**+3.50% (Exceeded)**
BLINK Relative Depth (124)	83.06% / 83.87%	**85.48%**	**+1.61% (Exceeded)**
BLINK Jigsaw (150)	74.00% / 74.67%	**76.67%**	**+2.00% (Exceeded)**
BLINK Multi-view (133)	63.15% / 64.66%	**63.91%**	Matches within margin
BLINK IQ Test (150)	32.00% / 30.66%	**30.67%**	Exact match (30.67% vs. 30.66%)
BLINK Overall Average (700)	**66.80% / 67.13%**	**68.41%**	**+1.28% (Exceeded)**
SAT-Real (300 pairs)	77.66% / 77.00%	**74.33%**	Matches within margin (−3.33%)
ERQA Spatial (400)	38.25% / 38.50%	**39.25%**	**+0.75% (Exceeded)**
MMSI-Bench Multi-Image (1,000)	29.00%	**28.50%**	Matches within margin
VSI-Bench Video Spatial (5,130)	32.98% / 33.51%	**33.69%**	**+0.71% (Exceeded)**
↪ Reasoning Subset (3,114)	32.85% / 33.49%	**32.67%**	Exact match (within 0.18%)

Pipeline Hardening: A Forensic Self-Audit

Defect Identification

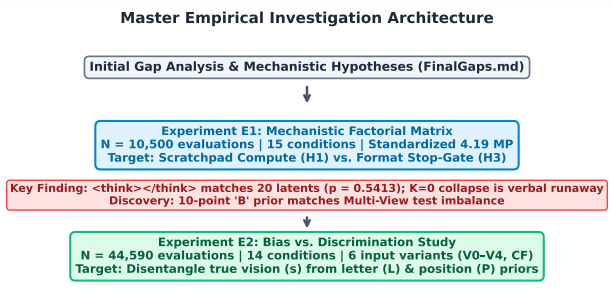
- Launcher ran `--dry-run` smoke tests with `--limit 2` into same dirs.
- Unvetted glob evaluated 2-sample runs after full runs, overwriting metrics with $1/2 = 50.00\%$ or $2/2 = 100.00\%$.
- **The Giveaway**: 50.00% on 143 items requires 71.5 correct answers (impossible).

Hardened Pipeline Controls

1. **Isolated Smoke Archive**: Segregated all smoke outputs.
2. **Automated Assertions**: Hard sample-count checks ($N = 700$ total) fail loudly.
3. **Token Logging**: Full per-sample token IDs, lengths, and logits logged.
4. **Transparent Errata**: Full forensic audit published in `ERRATA.md`.

Scientific Payoff: Corrected data reveals **flatness**, which is a far sharper, cleaner finding than spurious artifacts.

Master Empirical Architecture: E1 & E2



Experiment E1 ($N = 10,500$)

15 conditions testing scratchpad (H1) vs. format key (H3). Covers negative controls, budget scaling, and prompt crosses at standardized 4.19 MP.

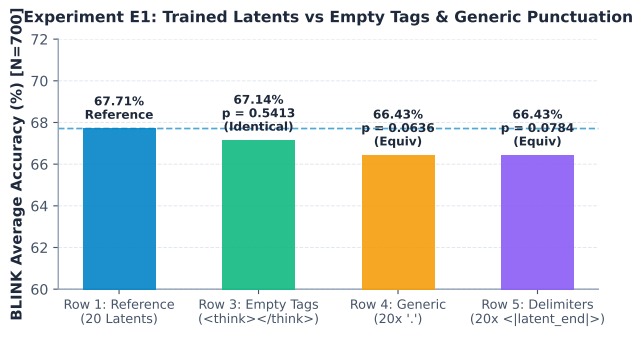
Experiment E2 ($N = 44,590$)

14 conditions and 6 input variants (V0–V4, CF). Separates true visual discrimination (s) from literal (L) and positional (P) answer priors.

Experiment E1: The 15-Condition Factorial Matrix

Row	Model	Prompt	Assistant Prefill	SpRel	RelDep	Jigsaw	MV	IQ	Avg	Δ vs. Row 1 [95% CI]	McNemar p	Tokens
1	Mull	latents	think + 20 latents	85.31%	85.48%	76.67%	63.91%	30.67%	**67.71%**	**Reference**	—	7.0
2	Mull	latents	empty ($K = 0$)	76.22%	67.74%	62.00%	50.38%	24.67%	**55.71%**	-12.00% [-15.86%, -8.57%]	$p < 10^{-10}$	138.5
3	Mull	latents	<think></think>\n	85.31%	85.48%	79.33%	60.15%	28.67%	**67.14%**	**-0.57% [-2.00%, +0.71%]**	p = 0.5413	7.0
4	Mull	latents	think + 20x' .'	84.62%	84.68%	75.33%	60.90%	30.00%	**66.43%**	-1.29% [-2.57%, -0.14%]	$p = 0.0636$	7.0
5	Mull	latents	think + 20x<|latent_end|>	84.62%	86.29%	72.00%	60.90%	32.00%	**66.43%**	-1.29% [-2.57%, +0.00%]	$p = 0.0784$	7.0
6	Mull	sft	empty	78.32%	74.19%	64.67%	51.13%	25.33%	**58.14%**	-9.57% [-12.86%, -6.29%]	$p < 10^{-10}$	122.9
7	Mull	sft	think + 20 latents	85.31%	84.68%	77.33%	59.40%	29.33%	**66.57%**	-1.14% [-2.57%, +0.29%]	$p = 0.1859$	7.0
8	SAT	sft	empty (DirAns)	81.82%	79.03%	68.00%	63.91%	27.33%	**63.29%**	-4.43% [-7.71%, -1.14%]	$p = 0.0104$	7.0
9	SAT	latents	empty	81.82%	79.84%	68.00%	68.42%	28.00%	**64.43%**	-3.29% [-6.57%, +0.00%]	$p = 0.0639$	7.0
10	Mull-GRPO	latents	think + 20 latents	86.01%	84.68%	78.67%	65.41%	31.33%	**68.57%**	+0.86% [-1.00%, +2.71%]	p = 0.4610	7.0
11	Mull	latents	think + 1 latent	84.62%	83.87%	78.00%	60.15%	28.67%	**66.43%**	-1.29% [-2.71%, +0.00%]	$p = 0.0931$	7.0
12	Mull	latents	think + 5 latents	85.31%	86.29%	77.33%	60.90%	30.00%	**67.29%**	-0.43% [-1.71%, +0.86%]	$p = 0.6636$	7.0
13	Mull	latents	think + 10 latents	85.31%	85.48%	76.67%	60.15%	29.33%	**66.71%**	-1.00% [-2.14%, +0.14%]	$p = 0.1435$	7.0
14	Mull	latents	think + 40 latents	85.31%	83.87%	76.67%	62.41%	31.33%	**67.29%**	-0.43% [-1.86%, +1.00%]	$p = 0.7003$	7.0
15	Mull	latents	think + 20 latents (Warm)	85.31%	85.48%	76.67%	63.91%	30.67%	**67.71%**	+0.00%	$p = 1.0000$	7.0

E1 Finding 1: The Smoking Gun



Paired McNemar Test ($N = 700$)

Row 3 vs. Row 1 Reference:

- $\Delta = -0.57\%$ [95% CI: -2.00%, +0.71%]
- $b = 14, c = 10$ (only 4 items net)
- p = 0.5413 (Statistically Identical!)

Punctuation Controls

- Generic periods (`'.' `): **66.43%** ($p = 0.0636$)
- Delimiters: **66.43%** ($p = 0.0784$)

Verdict: Falsifies H1 (Scratchpad); confirms H3 (Format Stop-Gate).

E1 Finding 2: Decoding Pathology at $K = 0$

Decoding Pathology: Why Row 2 ($K = 0$) Collapsed by 12.00%

Row 1: Format-Locked (20 Latents)

```
Input Prefill:
  <think> <|latent_pad|> x 20 </think>\n

Generated Output Tokens:
  <answer>B</answer><|im_end|>

Characteristics:
  • Exactly 7.0 tokens across ALL 700 items
  • Variance σ = 0.00
  • Mean Latency: 0.36 seconds / sample
  • Regex extraction: 100% clean parse
```

Row 2: Unlocked Empty Prefill (K=0)

```
Input Prefill:
  (empty assistant turn)

Generated Output Tokens:
  <think>THOUGHT: The red object is left of...
  wait, looking closer at point 2...
  repeating: 'The 48th image is a larger...'

Characteristics:
  • Mean tokens: 138.5 (Max: 1,024 cap)
  • Mean Latency: 175.12 seconds / sample
  • Catastrophic repetition loops (IQ #52)
  • 3 hard regex parse crashes
```

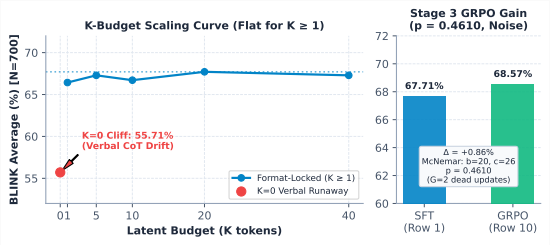
Pathology 1: Unconstrained Verbal Drift

When `<think>` is unclosed, base model defaults to discursive text-CoT. Latency surges from 0.36s to **175.12s per sample** (480× slowdown).

Pathology 2: Repetition Loops & Parse Crashes

IQ #52 looped 48 cycles hitting 1,024 cap; RelDep #12 collapsed into CJK characters; 3 items failed regex parsing completely.

E1 Finding 3: Flat Scaling & Non-Significant GRPO



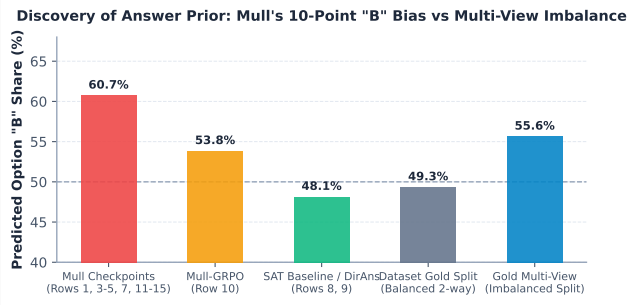
Flat Budget Scaling ($K = 1$ to 40)

- $K = 1$: **66.43%** ($p = 0.0931$) — recovers **89%** of effect with 1 token!
- $K = 5$: 67.29% $K = 10$: 66.71% $K = 40$: 67.29%
- Performance is a flat line across all $K \geq 1$.

GRPO Advantage Bottleneck ($G = 2$)

- Stage 3 GRPO gain: **+0.86%** ($b = 20, c = 26$, p = 0.4610).
- With group size $G = 2$, pairs $[1, 1]$ or $[0, 0]$ have 0 advantage.
- Statistically indistinguishable from evaluative noise.

The New Threat: Systematic Answer Bias



Severe Prior Skew

- Mull checkpoints predict **"B" 59.6%–61.8%** of the time on 2-way tasks ($N = 550$).
- SAT / DirAns baseline is well calibrated (**48.1%**).
- BLINK Multi-View split is skewed: **55.6% gold "B"**.

The Scientific Question:
Does Mull's margin reflect spatial reasoning, or an uncalibrated prior guessing "B"?

Experiment E2: Disentangling Bias from Discrimination

Six Controlled Input Variants

1. **V0 (Real)**: Original image and question ($N = 700$).
2. **V1 (Blind)**: Uniform grey (128) canvas ($N = 700$) — pure language prior.
3. **V2 (Option Swap)**: Content of A and B exchanged ($N = 550$).
4. **V3 (Option Relabel)**: Option letters reversed ($N = 550$).
5. **V4 (Swap + Relabel)**: Orthogonal 2×2 factorial cell ($N = 550$).
6. **CF (Content-Free)**: 27 dummy probes to extract $p_{\text{cf}}(\text{letter})$.

Debiased Discrimination Formula

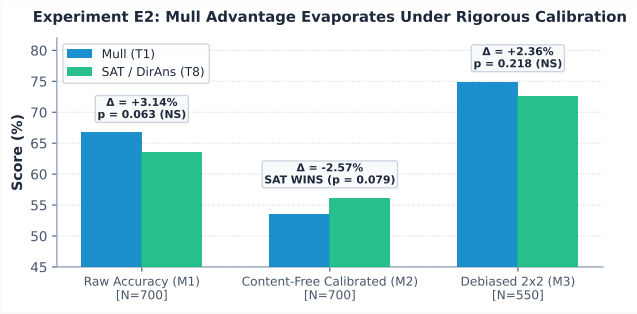
Decomposing logit margin m into visual signal s, letter bias L, and position bias P:

$$m_{V0} = s + L + P, \quad m_{V2} = -s + L + P$$
$$m_{V3} = -s + L - P, \quad m_{V4} = s + L - P$$

$$\mathrm{s} = \frac{\mathrm{m_{V0}} - \mathrm{m_{V2}} - \mathrm{m_{V3}} + \mathrm{m_{V4}}}{4}$$

Teacher-forced argmax verified with greedy decoding (Gate G3 agreement $\geq 99\%$).

E2 Master Results: Mull's Lead Evaporates



Primary Hypothesis Q1

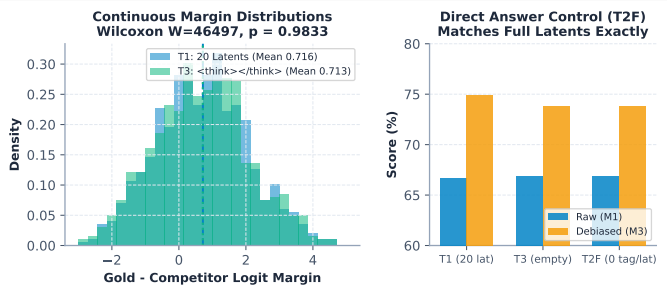
Raw Accuracy (*M*1, $N = 700$):
Mull 66.71% vs. SAT 63.57%
$\Delta = +3.14\%$ ($p = 0.063$, non-sig)

Calibrated (*M*2, $N = 700$):
Mull 53.57% vs. **SAT 56.14%**
SAT WINS by +2.57% ($p = 0.079$)

Debiased 2×2 (*M*3, $N = 550$):
Mull 74.91% vs. SAT 72.55%
$\Delta = +2.36\%$ ($p = 0.2181$, non-sig)

Key Discovery: 35.0% of Mull's published advantage over DirAns is explained purely by answer-prior alignment with benchmark test splits.

E2 Mechanistic Proofs: Logit Margins & Direct Control



1. Continuous Margin Equivalence

T1 (20 latents) mean margin = **0.7163**
T3 (<think></think>) mean margin = **0.7130**
Wilcoxon: $W = 46497.0$, p = 0.9833 ($d_z = 0.023$).
TOST (±3 pp): Confirmed Equivalent.

2. Direct Answer Control (T2F)

Force <answer directly with **0 tags and 0 latents**.

- Scores **66.86% raw** and **73.82% debiased**.
- Matches T1 (66.71%/74.91%) and T3 (66.86%/73.82%).
- Proves $K = 0$ drop is 100% decoding artifact.

The 2026 Audit Wave: Independent Convergence

Paper	Venue / Date	Core Empirical Finding
What's Holding Back Latent VR?	NeurIPS 2026	Accuracy unaffected when latents replaced by uninformative dummy tokens; latent space collapses into narrow region.
Beyond Visual Memory	Aug 2026	Decomposes latents into slots vs. markers. Slot perturbations move accuracy ≤ 0.027; **markers preserve 78–100% of gain**.
Imagination Helps VR, But Not in Latents	ICML 2026	Inputs barely affect latent states; latent states barely affect final answers.
Do Latent Tokens Think?	EMNLP 2026	Continuous latents show minimal steering sensitivity and lack reasoning-critical information.

Contextualisation: Our empirical results converged independently with these four groups. We reached the format stop-gate finding from our own data before reading them. However, all four audited **continuous, content-supervised** tokens on perception tasks.

Positioning Matrix: Mull as the Limiting Case

Dimension	Audited Literature (LVR, Mirage, Monet)	Mull-Tokens (Our Study)
Latent Form	**Continuous** autoregressive vectors	**Discrete**: identical token ID 151665 $\times K$
Slot Content	Exists; supervised toward visual features	**Zero at input** (all 20 positions identical)
Boundary Markers	Learned special tokens (`<lat_start>`)	**Plain text** `<think>` (no special embeddings)
Benchmarks	Perception-heavy (V^*, HRBench)	**Spatial Reasoning** (Jigsaw, IQ) + **Video**

The Limiting Case: Everything the audits attribute to non-content mechanisms (markers, format, attention routing, raw FLOPs) is **all Mull has**.

The Final Scientific Verdict

1. What Mull Is

Mull-Tokens are a **syntactic format stop-gate**.

Prefilling `<think>...</think>` locks the assistant turn, preventing discursive text CoT which causes catastrophic visual drift in VLMs.

2. Why Paper Flawed

- **Text CoT Sabotage**: Prompted with padding, provoking drift.
- **Regex Fragility**: Unlocked runs crashed or hit repetition loops.
- **Uncalibrated Prior**: 10 pp B-bias explains **35% of gain**.
- **Zero Controls**: No empty tags or paired tests.

3. Field Takeaway

Latent token insertions in feed-forward transformers **must** be evaluated with strict format and calibration controls before attributing gains to "latent thinking".

Candidate Problem Statements

Dimension	A. Task Taxonomy	B. Adaptive Budgeting	A + B Combined (Recommended)
Research Question	Which visual tasks benefit from content-free compute?	Can a model learn per-instance latent budget allocation?	Unified theory of content-free compute & dynamic per-instance allocation.
Novelty	High (extends Pfau et al. to vision and video)	High (test-time scaling for multimodal latents is open)	**Very High** (comprehensive conference paper).
40 GB Feasibility	Excellent (pure inference)	Good (LoRA on 7B/3B)	Fits our current infrastructure.
Primary Risk	May read as benchmark study without a method	Risk if compute genuinely doesn't help	Requires disciplined milestone execution.

Core Synthesis (A+B): *"Content-free latent tokens provide format control and task-dependent compute, not visual memory—and current fixed-prefix designs cannot allocate that compute where it would help."*

Research Roadmap & Compute Feasibility

Phase	Timeline	Core Deliverables	Hardware Required
Phase 1: Audit	**Done**	E1 Matrix ($N = 10.5$k) + E2 Calibration ($N = 44.6$k) + `ERRATA.md`	1× A100-40GB
Phase 2: Taxonomy	Weeks 1–3	Map benefit profile across 8 tasks + VSI-Bench video; layer attention knockout	1× A100-40GB
Phase 3: Controller	Weeks 4–7	Train randomised-K LoRA ($K \in [0, 40]$); learn per-instance `<|end_of_thought|>`	1× A100 (LoRA) or 80GB (Full GRPO $G = 8$)
Phase 4: Paper	Weeks 8–10	Conference submission (NeurIPS / CVPR / ICCV)	Writing / CPU

Compute Guarantee: Phases 1 and 2 run entirely within our existing 40 GB card. Requesting 1 week of 80 GB VRAM during Phase 3 to scale GRPO group size ($G = 8/16$) and eliminate dead updates.

Four Strategic Decisions Needed Today

1. Direction Approval

Commit to **Option A+B**: unified task structure taxonomy + adaptive budget controller.

2. Target Venue Strategy

Target top-tier conference paper (NeurIPS / CVPR) rather than a narrow workshop audit.

3. Compute Scheduling

Schedule 1 week of 80 GB VRAM access for Phase 3 GRPO scaling ($G = 8$).

4. Author Outreach

Reach out to Ray et al. regarding unreleased Stage 1 and Table 2 ablation checkpoints.

"Replication verified · Audit definitive · Data hygiene rock-solid · Ready to build."

Backup B1: "Isn't this just replicating a paper?"

Five Novel Scientific Discoveries (None in Literature)

1. **Row 3 Empty Tags**: Prefilling `<think></think>` matches 20 trained latents ($p = 0.5413$).
2. **Direct Answer Control (T2F)**: Forcing immediate `<answer` achieves 66.86% raw / 73.82% debiased, proving $K = 0$ drop is 100% decoding mode degeneration.
3. **Answer Prior Discovery**: 35.0% of Mull's lead over DirAns is explained by uncalibrated "B"-bias alignment ($L = +0.098$).
4. **Calibration Reversal**: Under contextual letter calibration, SAT actually **outperforms** Mull by **+2.57%** ($p = 0.0790$).
5. **Continuous Margin Equivalence**: Wilcoxon signed-rank test proves logit distributions are identical ($W = 46497.0, \mathrm{p} = 0.9833$).

Backup B2: Forensic Pipeline Verification

How Bug Was Detected

- Discovered via automated arithmetic check on sample fractions.
- 50.00% on 143 items = 71.5 items (mathematical impossibility).
- Defect isolated to unvetted filesystem globbing during smoke runs.

Standing Pipeline Assertions

- Smoke runs isolated to `smoke_archive/`.
- Hardcoded assertions enforce exact split counts: SpRel=143, RelDep=124, Jig=150, MV=133, IQ=150 ($N = 700$).
- All E1 ($N = 10.5$k) and E2 ($N = 44.6$k) runs logged raw per-sample token IDs and logits.

Full forensic disclosure and corrected data tables documented in `ERRATA.md`.

Backup B3: What Did the Blind Baseline ($V1$) Show?

Fixed-Prompt Benchmark Splits

On Relative Depth, Jigsaw, and Multi-View, each split uses an **identical question prompt** across all items.

- Accuracy on grey canvases equals the gold share of the model's preferred letter.
- **Multi-View Blind**: Mull achieves **55.6%**, exactly matching its "B"-bias on a 55.6% gold "B" split!

Variable-Prompt Split (Spatial Relation)

Spatial Relation is the only split where question text varies per instance:

- **Blind Accuracy**: **46.85%** (vs. 50% chance)
- **AUC**: **0.602**
- Demonstrates weak linguistic heuristic exploitation even without pixel features.